INCH-POUND

MIL-DTL-32096A(OS)
29 October 2002
SUPERSEDING
MIL-DTL-32096(OS)
6 May 2002

DETAIL SPECIFICATION

CARTRIDGE CASE, STEEL, 5-INCH, MK 9 MOD 3 (54 OR 62 CALIBER)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the 5-inch, 54- or 62-caliber steel cartridge case, herein referred to as the "cartridge case."

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1684 Tape, Pressure-Sensitive Adhesive (Plastic Film)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Documentation Branch (Code 4230), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1320

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DEPARTMENT OF DEFENSE

MIL-DTL-117	Bags, Heat Sealable
MIL-S-3289	Steel, Plate and Disk, Carbon, Forging Quality
MIL-D-3464	Desiccants Activated, Bagged, Packaging Use and Static Dehumidification

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-129	Marking for Shipment and Storage
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Codes 10001 and 53711)

444048	5in/54 Caliber Case Body Gage Assembly, Receiver
3208375	Charge, Propelling, 5in/54 Caliber, Mk 67 Mod 3
7263988	Case, Cartridge, MK 9 Mod 3

(Application for copies of NAVSEA drawings should be addressed to: Commander, Naval Surface Warfare Center, Crane Division (Attn: Code 4021), Crane, IN 47522-5001.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OD 47471	Method for Rating the Effects of Salt Spray Testing on Zinc Electroplated Cartridge Cases

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Guns Branch (Code 4210K), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted

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are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117	Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD adopted)
ASTM B 499	Standard Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals (DoD adopted)
ASTM B 633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel (DoD adopted)
ASTM D 5486	Standard Specification for Pressure-Sensitive Tape for Packaging, Box Closure, and Sealing (DoD adopted)
ASTM E 8	Standard Test Methods for Tension Testing of Metallic Materials (DoD adopted)
ASTM E 18	Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(Application for copies should be addressed to the American Society for Quality, P.O. Box 3005, 611 E Wisconsin Ave, Milwaukee, WI 53201-3005.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Characteristics. Unless otherwise specified in the contract or purchase order, the cartridge case shall conform to the requirements specified herein and on Drawing 7263988 (see 4.5.1).

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3.3 High stress expansion (M101). The cartridge case shall show no evidence of cracks in any portion of the case when subjected to the high stress expansion test of 4.5.2.

3.4 Hardness (M102). The cartridge cases shall conform to the hardness value for region D of figure 1 shall be as specified in table I. Values for regions A1, A2, B, and C, as shown in figure 1, are advisory (see 4.5.3).

3.5 Tensile strength and elongation (M103). The yield strength and elongation values for the cartridge case shall be as specified in table I in the regions shown in figure 1 (see 4.5.4 and 4.5.5).

TABLE I. Hardness and tensile strength.

Region	Hardness[1]		Yield Strength at 2% offset, (psi)	Elongation (1" gage length)
	Max[2]	Min		
A1	C42	B75	---	---
A2	C42	B90	---	---
B	C42	C34	135,000 Min	5% Min
C	C42	B95	90,000 Min	6% Min
D	B85	--	---	---

[1] Rockwell "C" or "B" scale.

[2] Stress relief and hydrogen embrittlement of ASTM B 633 do not apply.

3.6 Protective coating (M104). The case shall be zinc plated in accordance with ASTM B 633, Type II except that the exterior thickness of the protective coating shall be between 0.0003 inch and 0.0008 inch and the interior plating thickness shall be 0.00015 inch minimum. This requirement does not apply within one inch of the head. Thinner plating at the inert standoff contact points is acceptable provided there are no more than nine. The protective coating shall show no corrosion of basis metal. The size and frequency of the black corrosion products of zinc shall not exceed that of the established standard of 0.2400 (see 4.5.6). After testing, white corrosion products of zinc that do not interfere with chambering of the cartridge case are acceptable.

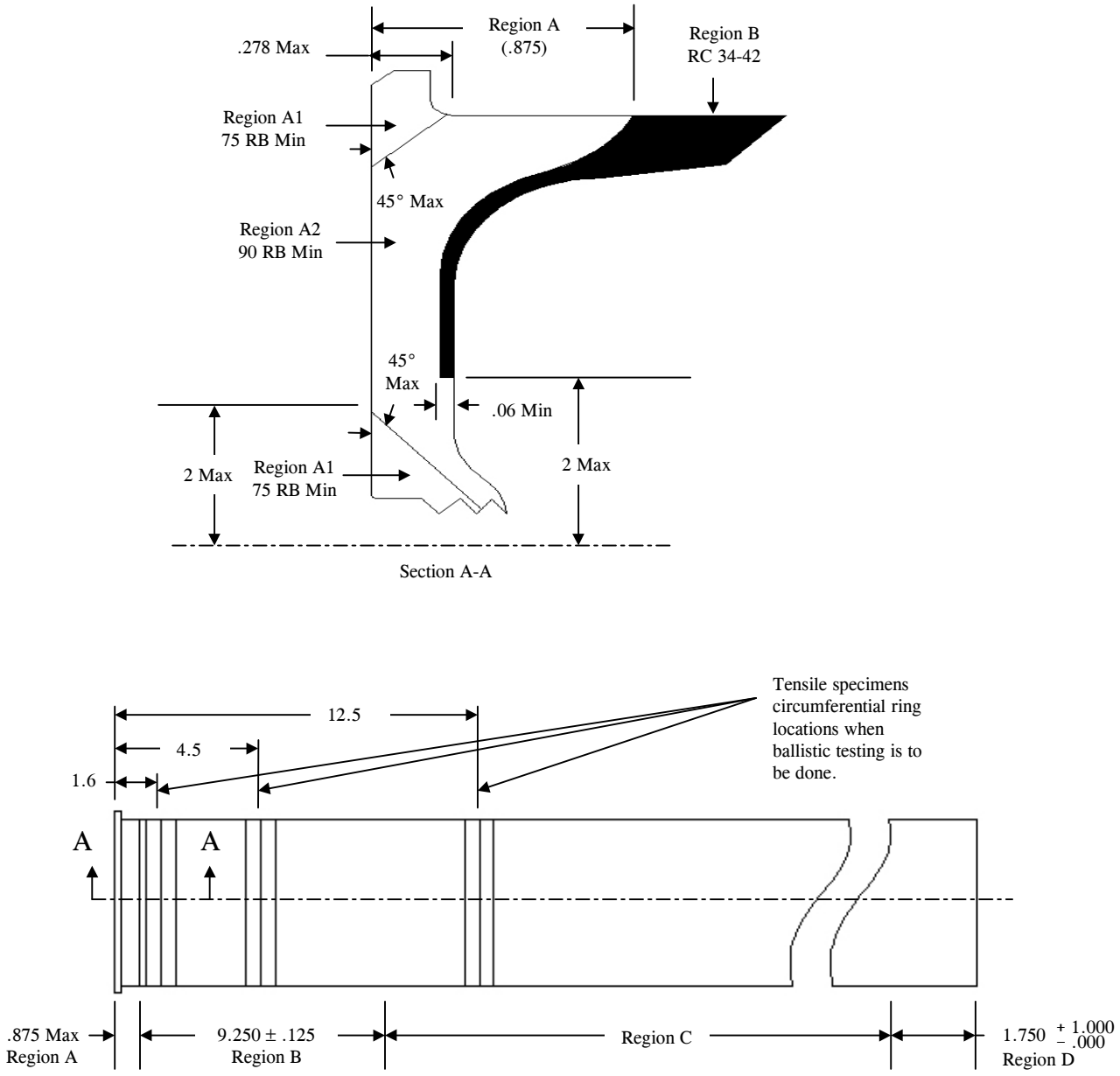
3.7 Chamber gage (M105). Each cartridge case shall fully chamber in a Navy-approved chamber gage (see 4.5.7).

3.8 Ballistics (M106). Each cartridge case shall load, eject, show no splits, and otherwise perform satisfactorily when ballistically tested (see 4.5.8).

3.8.1 Case extraction (M107). Cartridge cases tested under automatic and semiautomatic action shall be ejected and shall in no way cause the weapon to cease firing. When fired in the single shot mode, the cartridge case shall be capable of being extracted by manual operation of the extraction system.

Note: The term "ejected" means that the cartridge case is thrown clear of the breech, permitting loading of the subsequent round. The term "extract" means that the cartridge case is loosened and pulled to the rear, by the normal operation of the extraction system, sufficiently to permit ready removal.

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Note: Dimensions are in inches.

FIGURE 1. Tensile and hardness specimen locations.

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3.9 Item identification. Item identification marking of all cartridge cases shall be in accordance with the contract or purchase order (see 6.2).

3.10 Workmanship (M108). The cartridge case shall be produced in such a manner as to ensure compliance with all the requirements of this specification and the applicable associated specifications. There shall be no burrs, cracks, tears or other imperfections of manufacture, which in the opinion of the Government inspector, may adversely affect either the assembly, functioning, or handling of the product. (Flutes, draw marks, loopers, slug marks, and pits shall not exceed 10 percent of the calculated wall thickness in the affected area nor be of sufficient magnitude to cause minimum or maximum dimensions to be exceeded.

Note: Grain orientation directionality is produced by rolling. A looper is a visible pattern of this grain orientation extended by drawing.

There shall be no flutes, pitting, draw marks or loopers in the region between the base and a circumferential line 3 inches from the base of the cartridge case.) The only acceptable surface defects on the interior base-to-wall radius are those exemplified by the sample board approved by the contracting activity (see 4.5.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Classification of characteristics. The characteristics verified by the tests and examinations specified herein and in the detail inspection are classified as critical, major or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements which are not annotated with a classification code are classified as minor.

4.3 First article inspection. The first article sample shall consist of 158 cartridge cases and shall be examined and tested in accordance with table II. The cartridge cases shall be the same as for regular production and manufactured under the same conditions as the production lot. Further production of cartridge cases by the contractor, prior to the approval of the first article, shall be at the contractor's risk.

4.3.1 First article sample acceptance/rejection. Acceptance or rejection of the first article sample shall be in accordance with table II. For those where the acceptance criteria is not shown, any failure to meet the requirements shall be cause for rejection of the sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests as specified in table II.

4.4.1 Lot size. Unless otherwise specified in the contract or purchase order (see 6.2), the production lot shall consist of 10,000 cartridge cases plus those required for test purposes.

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TABLE II. First article sample and quality conformance inspection and sampling.

Test Or Examination	Requirement Paragraph	Method Paragraph	First Article Sample Size	Quality Conformance Inspection	
				Test Units Per Lot	Consequences Of Failures
Dimensional	3.2	4.5.1	158	56	Reject lot
High stress expansion	3.3	4.5.2	158	100%	Reject defective units
Hardness	3.4	4.5.3	5	5[1]	Reject lot
Tensile strength and elongation	3.5	4.5.4 and 4.5.5	5	5[1]	Reject lot
Protective coating a. Thickness b. Corrosion	3.6	4.5.6	 3 3	 3 1[2]	 Reject lot Reject lot[2]
Chamber gage	3.7	4.5.7	158	100%	Reject defective units
Ballistic	3.8[3]	4.5.8	150	50	Reject lot
Item identification	3.9	4.5.1	158	100%	Reject defective units
Workmanship	3.10	4.5.1	158	100%	Reject defective units
Packaging and marking	Section 5	4.6	158	100%	Reject defective units

[1] Unless otherwise specified in the applicable associated specification.

[2] Nonconformance of one sample to meet the corrosion code requirements shall require testing of an additional 3 samples. Failure of any of the additional samples to meet the corrosion code requirement shall be cause for rejection of the lot.

[3] Nonconformance of one sample to meet the case ejection requirement of 3.8.1 shall require testing of an additional 100 samples. Failure of any of the additional samples to meet the ejection requirement shall be cause for rejection of the lot.

4.5 Methods of inspection.

4.5.1 Visual and dimensional examination. The charges shall be visually and dimensionally examined for conformance to requirements of 3.2, 3.9 and 3.10. Four-power (4X) magnification may be used for examination of defects.

4.5.2 High stress expansion test. Each cartridge case in the lot (see 4.4.1), after heat treatment but prior to tapering, shall be subjected to a high stress expansion test. The cartridge case shall be expanded diametrically 0.050-inch minimum while under pressure, in a zone extending from 1.5 inches above the

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flange to 9 inches above the flange. Expansion will decrease as the flange is approached with expansion at the flange to be zero.

4.5.3 Hardness. Cartridge cases shall be prepared and tested in accordance with ASTM E 18 to determine conformance to 3.7. The hardness shall be checked at the locations shown in Figure 1. Hardness readings shall be taken on each of two sections cut longitudinally from diametrically opposite quadrants of each cartridge case.

4.5.4 Elongation test. Cartridge cases shall be tested in accordance with ASTM E 8 to determine conformance to 3.8. The case shall be sectioned circumferentially as specified in the applicable associated specification. Specimen dimensions shall be as indicated in figure 6 of ASTM E 8. Specimens shall be subsize specimens or as directed in the contract (see 6.2). Unless otherwise specified, the specimens shall be straightened by pressing and rolling and shall be stress relieved at 610°± 10° F for 30 ± 5 minutes and then cooled at room temperature for a minimum of 1 hour. The head separation rate shall be 0.05 inch per minute.

4.5.5 Tension test. Circumferential specimens cut from three positions shown in figure 1 shall be tested for yield strength in accordance with ASTM E8. An offset value of 0.002 inch per inch length shall be used to determine yield strength.

4.5.6 Protective coating tests. Unless otherwise specified in the contract or purchase order (see 6.2) or applicable drawings, nondestructive measurement of coating thicknesses shall be made using the procedure specified in ASTM B 499. Measurements of interior coating thickness may be made on samples taken for hardness testing. Unless otherwise specified in the contract or purchase order (see 6.2) or applicable drawings, the protective coating shall be tested in accordance with ASTM B 117 with continuous exposure to salt spray for 96 hours. The black speck corrosion rating procedure shall be as specified in OD 47471.

4.5.7 Chamber gage examination. Each cartridge case shall be chamber-gaged in receiver, Drawing 444048, to determine conformance to the requirements of 3.7. A gage removal force of 59 pounds maximum is permitted.

4.5.8 Ballistic test. The cartridge cases shall be loaded in accordance with Drawing 3208375, using the assessed charge weight for the propellant lot being used. Projectile weight shall be adjusted to yield operating pressure of 65,000 psi and proof pressure of 75,000 psi at ambient temperature. (Charge and projectile weights will not be adjusted for firings at temperature extremes.) Testing shall be performed in a Mark 45, Mod 4 gun mount and in quantities specified in table III.

4.5.8.1 Test procedure. All proof pressure rounds shall contain M11 copper crusher gages, and shall be fired in the single shot mode to determine compliance with the pressure requirements. The remainder of the cartridges shall be fired in the automatic mode. Temperature of the rounds shall be as specified in table III. Rounds shall be temperature conditioned for a minimum of 24 hours.

4.6 Inspection of packaging. Packing and marking shall be visually examined for conformance to section 5.

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TABLE III. Ballistic acceptance.

Type of lot	Number of round	Type of round	Round temperature °F
First article	70 30 25 25	Service Proof Service Service	70 70 120 20
Production	40 10	Service Proof	70 70

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES (This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The Mark 9 Mod 3 steel cartridge cases, when assembled with their appropriate primers, plugs and propellant charges form complete charges which are used to fire the family of projectiles available for the Mk 45 gun mounts. The Mark 9 Mod 3 steel cartridge cases are designed for military use and have no equivalent commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Whether a first article is required (see 3.1).

d. Agency responsible for inspection

e. Production lot size, if other than as specified in 4.4.1.

f. Measurement and testing of coating if different from 4.5.6.

g. Packaging and special markings required (see 5 and 6.5).

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h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M, are applicable (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Federal Acquisition Regulations (FAR) 23.3.

Note: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.5 Packaging and Marking.

6.5.1 Packing. When specified by the contracting activity (see 6.2), cartridge cases should be packed by the contractor for domestic shipment in 200 lb test RSC boxes conforming to Rule 41 of Uniform Freight Classification UFC 6000**.** Care should be taken to protect the mouth and the primer hole and to separate the cases so none will be damaged in shipment**.** Cartridge cases should be packed in each box with the mouth of the cartridge case in an upright position**.** The boxes should be appropriately marked to assure that the mouth of the cartridge case will be maintained in an upright position. Packing should be four cases packed in each box. The boxes should have full length partitions between the cases with flaps at the bottom to fold over and provide a bottom pad. A top pad will fit around the mouths of the cases to prevent them from striking one another in the box and damaging their mouths. Each case is to be packaged in a bag in accordance with MIL-P-117, Type III, Class E, Style 1. Approximate size of bag is 11 ½" by 43" inches. Place a one unit bag of desiccant, per MIL-D-3464, Type I, inside before sealing. Close bag and seal with tape conforming to A-A-1684. Tape conforming to ASTM D 5486 may be used as an alternate tape.

6.5.2 Palletizing. Cases packed in boxes should be palletized in accordance with standard commercial practice.

6.5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), all marking should be in accordance with MIL-STD-129. An arrow should be stenciled on each container as specified in MIL-STD-129.

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6.6 Subject term (key word) listing.

Ballistic cases
Chamber gage

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Preparing Activity:
Navy – OS
Project 1320-0051

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:

1. DOCUMENT NUMBER: **MIL-DTL-32096A(OS)**

2. DOCUMENT DATE (YYMMDD): **022902**

3. DOCUMENT TITLE: **CARTRIDGE CASE, STEEL, 5-INCH, MK 9 MOD 3 (54- OR 62-CALIBER)**

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME
Commander, Indian Head Division
Naval Surface Warfare Center

b. TELEPHONE (*Include Area Code*)
(1) Commercial: 301-744-6554
(2) DSN: 354-4700

c. ADDRESS (*Include Zip Code*)
Documentation Branch (Code 4230)
101 Strauss Avenue
Indian Head, MD 20640-5035

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